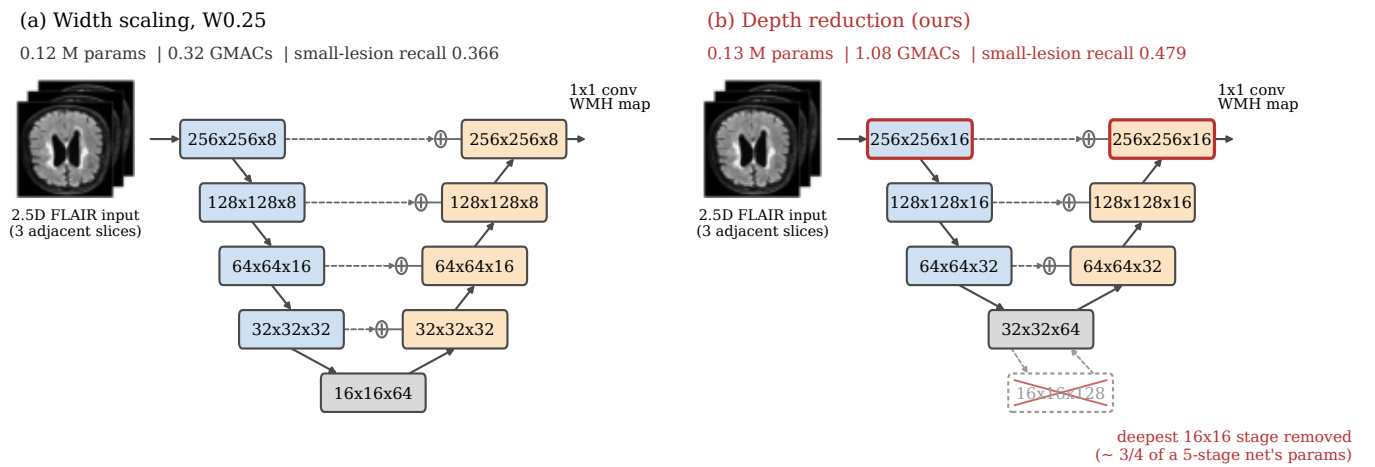
(a) Width scaling, W0.25
0.12 M params | 0.32 GMACs | small-lesion recall 0.366
2.5D FLAIR input
(3 adjacent slices)
256x256x8
256x256x8
128x128x8
128x128x8
64x64x16
64x64x16
32x32x32
32x32x32
16x16x64
1x1 conv
WMH map
all five stages kept;
every stage narrowed to 0.25x the baseline
encoder block: 2x (conv 3x3 - IN - LeakyReLU)
decoder block: up-conv, concat, 2x conv
(b) Depth reduction (ours)
0.13 M params | 1.08 GMACs | small-lesion recall 0.479
2.5D FLAIR input
(3 adjacent slices)
256x256x16
256x256x16
128x128x16
128x128x16
64x64x32
64x64x32
32x32x64
16x16x128
1x1 conv
WMH map
deepest 16x16 stage removed
(~ 3/4 of a 5-stage net's params)
saved budget moved to the high-resolution
stages (2x the channels of W0.25)
bottleneck (deepest stage)
skip connection (channel concatenation)